

Solution Architecture

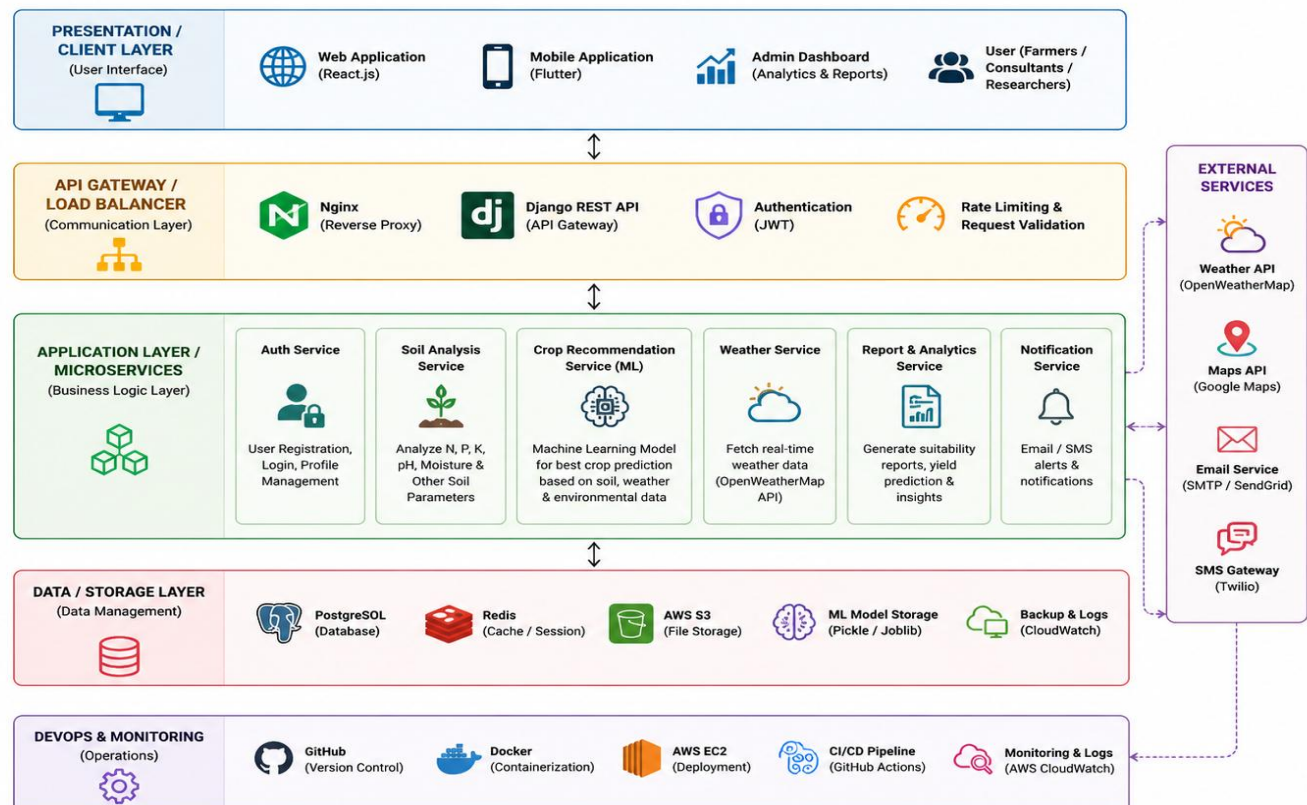
Date	28 June 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

SOLUTION ARCHITECTURE

Smart Agricultural Production Optimization Engine (OptiCrop)



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer (Client Layer)	Provides user interface for farmers to input soil & environmental data, view crop recommendations, suitability reports, and dashboard analytics.	React.js, HTML5, CSS3, Bootstrap, JavaScript, Chart.js
API Gateway (Communication Layer)	Handles all client requests, routes them to appropriate services, ensures security, rate limiting, and request validation.	Django REST Framework, Nginx (Reverse Proxy), JWT Authentication
Microservices (Application Layer)	Contains core business logic and machine learning services. Processes data, makes predictions, manages users, and integrates external services.	• Auth Service – User registration & login • Crop Recommendation Service – ML based crop prediction • Soil Analysis Service – Soil health evaluation • Weather Service – Real-time weather data • Report Service – Generate reports & analytics Technologies: Python, Django, Scikit-learn, Pandas, NumPy, Celery (for tasks)
Database (Data Layer)	Stores all application data including users, soil & weather data, crop datasets, prediction results, and reports.	PostgreSQL (Primary Database), Redis (Caching / Session Management), AWS S3 (File Storage)